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Too hot to handle?

The Olympic Games in Paris may be the hottest yet, creating a dangerous environment for competitors, says **Madeleine Orr**

THE last summer Olympics, in Tokyo, were the hottest on record. The Paris games, kicking off next week, could be hotter. With wide boulevards, busy roads and zinc-topped roofs, Paris is a heat island. Daytime high temperatures have routinely topped 30°C in July and August in recent years. Add high humidity and competitions scheduled around midday, and it becomes a potentially dangerous environment for competitors.

The heat risk experienced by athletes is different to that for those of us watching from the sidelines. Exercise produces considerable warmth in the body as a by-product of movement. That heat is dissipated through the skin, which becomes red and flushed. This happens both via radiative losses and, vitally, via sweat, which evaporates to cool us. In summer, the temperature gradient with our surroundings narrows, so radiative cooling declines or stops. If rising temperature, high humidity and a lack of a breeze make it impossible for the body to sweat properly, it can spell big trouble.

This state occurs at a “feels-like” temperature in the mid-30s and above. In these conditions, heat can’t escape the body and continues to build. Initially, breathlessness, fatigue, some cramping, maybe some nausea and light-headedness occur. Dehydration will make any of these symptoms worse.

If cooling interventions aren’t provided, and body temperature



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doesn’t drop, catastrophic outcomes like fatal heat stroke can follow with just 30 minutes of additional exposure. Several Olympic events last long enough for that to happen.

A report called *Rings of Fire* from the British Association for Sustainable Sport highlights the risks for different disciplines, including track and field, marathon swimming, tennis, hockey, rugby and cycling. This isn’t new information. In the US, the National Center for Catastrophic Sport Injury Research has been tracking fatalities linked to exertional heat strokes since the

1980s and has recorded more than 60 deaths. Many more are assumed to have occurred, but aren’t in the figures due to a lack of awareness and understanding among trainers, coaches and the families of those affected.

It remains unclear whether Olympic organisers are heeding the warnings. Some adaptive measures are in place, like cooling baths, hydration units, misting fans and shaded staging areas. But it may not be enough.

While organisers initially chose to forego air conditioners by opting for an energy-efficient geothermal water-cooled set-up

in the Olympic village, some national teams elected to bring their own, and many others complained. Ultimately, the hosts caved and ordered 2500 AC units that will be installed before the competitors arrive.

I don’t think any athletes will die at these Olympics because the medical provision is so good at such events, and the reputation risk would be so high that organisers will do everything in their power to prevent a catastrophic outcome. Still, if conditions do get hot enough, it could happen. The early symptoms of heat stress aren’t visible to others. Athletes will have to self-report their symptoms, and many who reach the Olympics will be reluctant to compromise their chance to compete.

What worries me most about conditions at these games isn’t the plight of competitors. It is that of young, impressionable athletes around the world and their well-meaning coaches and parents who may watch Olympians continue in unsafe conditions and think it is OK to do the same, but without all the resources, cooling facilities and medical support on hand.

Hosting an Olympic Games in Paris was a good idea. Scheduling them for July and August wasn’t. ■



Madeleine Orr is the author of *Warming Up: How climate change is changing sport*